USER GUIDE FR3D MK3

1. Project objective

Build a filament extrusion and spooling platform with digital control, telemetry and traceability, based on the MK3S+ with FR3D Addon firmware operating in standalone mode from the machine LCD.

Documentation scope. This document is directed exclusively at the MK3S+ with FR3D Addon operating in standalone mode (LCD and onboard firmware control). It does not cover Raspberry Pi gateway pairing or the browser-based operations UI.

Diameter sensor hardware. Hardware guidance here is based on a Hall-effect filament diameter sensor connected to pin A3 on the MK3S+ PCB. The same FR3D Addon setup can also operate with the optical filament diameter sensor built by ARTME 3D.

2. Addon FR3D

- Modified MK3S+ firmware

2.1 Diameter Meter Caliper

The Caliper is only for calibrating the diameter gauge and ensuring it measures correctly.

Calibration is only done once during assembly.

Then, the offset is used to adjust the zero point if it's misaligned (as is the case with all

calibrators on the market that have a zero adjustment).

Specifically, this function is only for calibrating the gauge so that the display shows the actual

measurement, where the lower limit is 1.5 mm and the upper limit is 2.5 mm.

The gauge will always indicate the actual measured value between 1.5 and 2.5 mm. Outside

these limits, it displays 1.5 or 2.5 mm due to a mechanical issue related to the physical sensor.

2.2 Diameter Predictor (FR3D MK3S+)

2.2.1) What is the Predictor?

The Diameter Predictor is a closed-loop assistant that runs on

the MK3S+ FR3D firmware.

It uses the Hall filament sensor calibrated in the previous step to keep the extruded diameter

close to the target (typically 1.75 mm below one of the parameters given to the predictor).

Every ~10 seconds, the predictor analyzes the data and, if correction is required and safety

conditions are met, the algorithm adjusts:

1) Extruder speed (E): This is the fastest response to diameter errors.

2) Nozzle temperature (T): When RPM alone is insufficient, the predictor switches to

a thermal correction strategy.

2.2.2) Correction (Safety Gates)

The predictor does not act unless, among other checks:

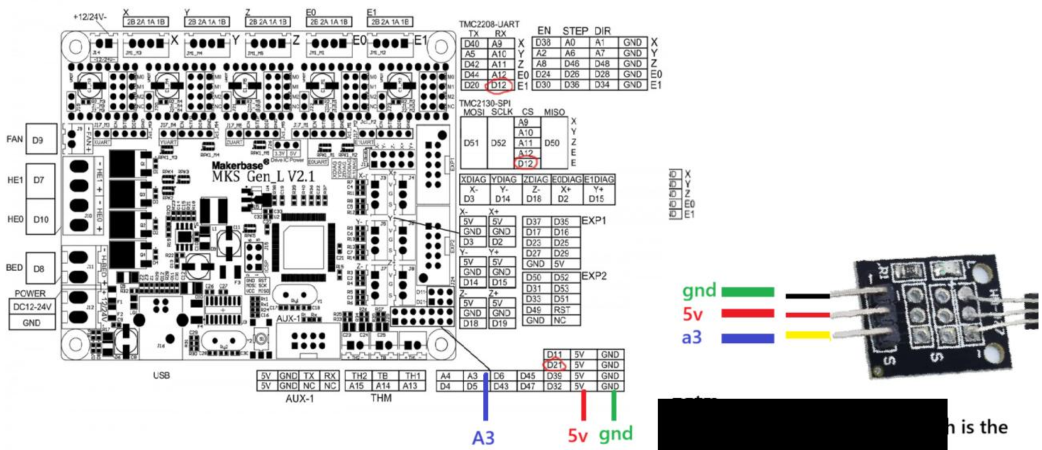
- The predictor is in automatic mode.
- The extruder is running (ES = 1).
- The current temperature is close to the setpoint (|Tact − Ttgt| ≤ the maximum configured
- temperature).
- Spolling is set to automatic.
- The diameter signal is stable, with no electrical noise (medium debounce: large jumps
- require confirmation using Med jump/Med confirm).
- The proposed R and T values are maintained within Rmin/Rmax and Tmin/Tmax. (This

  Determines the minimum and maximum adjustment limits of MK3S+.)

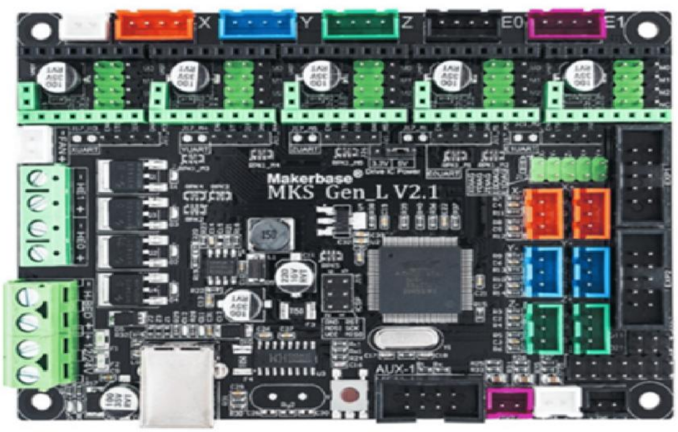
- What is the target diameter and what margin of error does it tolerate before taking action?

3. Diameter Hardware connection

Recommended Technical Note: Noise Reduction Filter for the SS495A Hall Effect Sensor. To protect the highly sensitive SS495A linear Hall effect sensor from high-frequency electromagnetic interference (EMI) caused by stepper motors and heating elements, a two-stage filtering strategy is implemented. This will be done in two stages: one for decoupling the power supply and another RC low-pass filter stage for the analog signal.



[redacted]



Stage 1 - Power Supply Decoupling (located 4 cm from the sensor). Components:

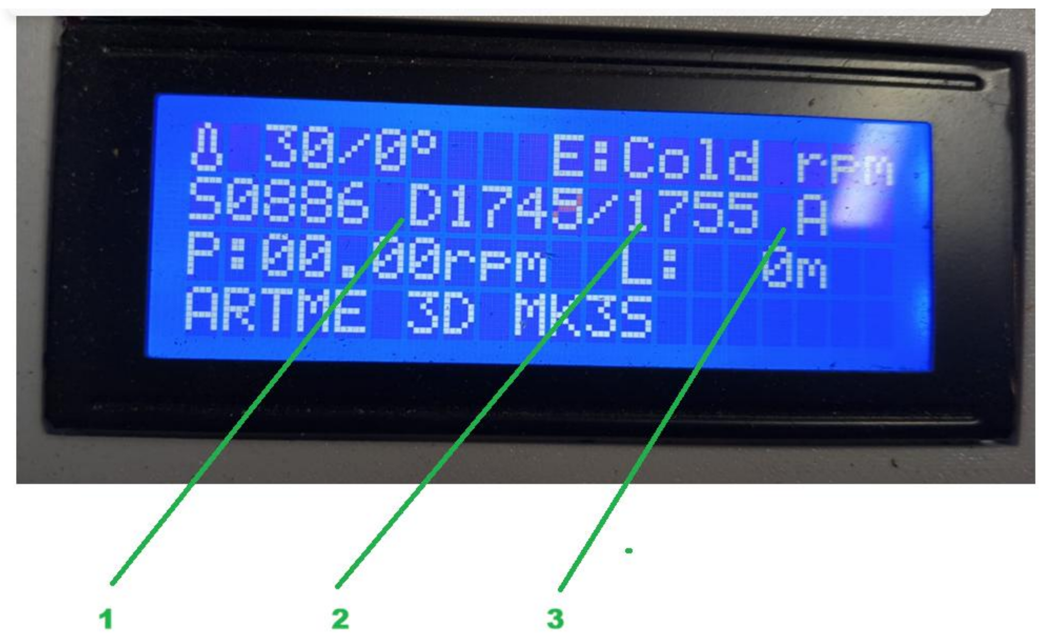
- One 0.1 µF ceramic capacitor (labeled 104)
- One 4.7 µF electrolytic capacitor.

Wiring Setup: Connect the positive (+) lead of the 4.7µF electrolytic capacitor (the longer leg) to one leg of the 0.1µF ceramic capacitor. Solder this connection directly to the 5V (Vin) line. Connect the negative (-) lead of the electrolytic capacitor (the shorter leg aligned with the strip on the body) to the remaining leg of the ceramic capacitor. Solder this connection directly to the GND line. Note: The sensor's output (data) wire completely bypasses this block and continues along the line. Stage 2 - Signal Data Filtering (RC Low-Pass Filter): Components:

- One 1kΩ resistor
- One 10nF ceramic capacitor (labeled 103).

Wiring Setup: Resistor Placement: Cut the signal (output) wire coming from the sensor. Solder the incoming wire to one end of the 1kΩ resistor. Connect the other end of the resistor directly to the microcontroller's analog input pin. Capacitor Placement (Critical): Place the 10nF ceramic capacitor as close as possible to the microcontroller pin (ideally less than 4 cm or directly on the board). Solder one leg of the capacitor to the analog input pin (the same junction where the resistor ends) and the other leg directly to the microcontroller's GND pin.

4.MK3S + FR3D main screen



(1) Displays current filament diameter

(2) Displays average filament diameter (from the last 10 seconds - with 20 samples)

(3) Displays prediction mode: A = automatic adjustment, M = manual adjustment

4.1 Configuration

You can configure from Pyhton or MK3 menu on Control AddonFR3d

4.1.2 MK3s HallA3 setting

Go to Menu-Control-AddonFR3D-HallA3

Setting value of these 3 reference diameters, you need a pattern for 1.7, 1.75 and 1.8 mm to

calibrate its.

4.1.2.1 MK3s Diameter Meter Calibration:

Step 1) Insert the 1.75 mm filament into the Hall sensor hole.

Step 2) Go to Menu > FR3D Plugin > HallA3 > Capture 1.75

Step 3) Wait at least 5 seconds for a stable analog reading.Step 4) When the reading is stable, click the encoder button and the value will be saved.

Step 5) Repeat steps 1 through 4 for Capture 1.7 and Capture 1.8, using 1.7 mm and 1.8 mm

patterns, respectively.

4.1.2.2 Calibration Verification and Adjustment

1) Insert the 1.75 mm filament into the Hall sensor hole.

2) Wait at least 15 seconds and go to the main menu. On the second row of the LCD screen, you

will see the diameter value DXXXX/YYYY. XXXX is the actual measurement, and YYYY is the

average measurement in mm. It should display 1750.

4.1.2.3 OFFSET Compensation Adjustment

If the value on the screen is not 1750, but, for example, 1720, you can adjust the compensation

as follows:

Step 1) Go to Menu - FR3D Plugin - A3 Panel - Compensation

Step 2) Calculate the difference, for example: 1750 - 1720 = 0.030

Enter the value 0.030 in Compensation and press the Click button.

Step 3) Check again on the main screen that it now displays D1750/1750.

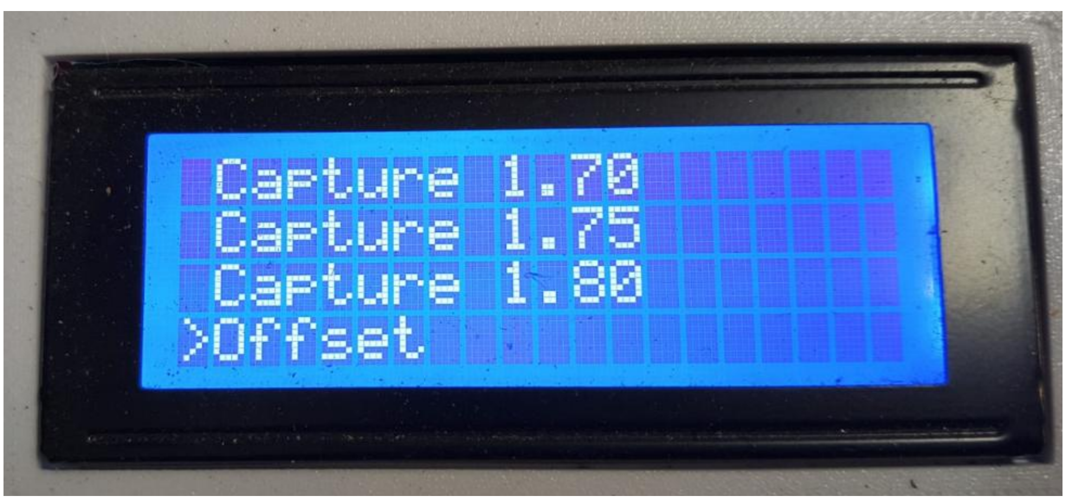
Note: This compensation adjustment only needs to be performed each time you want to check

the caliper.

Below you can see lcd screenshot

Menu AddonFR3d

Menu Capture Menu Capture 1.75 (without 1.75 pattern) Menu Capture 1.75 (with 1.75 pattern)



4.2 MK3 Predictor Diam


Tmax: 189
Deadband/2: 000.03
Target D mm:001.75
>Advanced →


Predictor
E margin: 002.00
T margin: 2
|T-Ttgt| ma:002.00


T settle N: 3
Med jump mm:000.15
Med conf mm:000.06
>DIAMDEBUG ON →

Emin — Lower RPM limit allowed for extrusion-speed corrections. Prevents the predictor from reducing speed too much and compromising flow continuity.

Emax — Upper RPM limit allowed for speed corrections. Prevents over-acceleration and saturation-driven oscillations in the extruder.

Tmin — Lower thermal limit of the assistant working range (it does not change the base RTF setpoint by itself). Used to bound temperature correction decisions.**Tmax** — Upper thermal limit of the assistant range. Keeps corrections inside a safe zone for material/process.

Deadband/2 mm — Half of the deadband around target diameter. If diameter stays within target ± band, predictor does not correct. Larger bands = fewer corrections; smaller bands = more sensitivity.

Target D mm: — Target filament diameter the system tries to maintain. All adjustment and deadband logic is computed against this value.

=====**Advanced**=====

Advanced-Wait after T change — Number of cycles/fusions the predictor waits after applying a temperature change before correcting T again. Reduces thermal-inertia rebound.

Advanced-E margin to switch to T (rpm) — Hysteresis that authorizes transition from RPM correction to temperature correction. Avoids overly frequent strategy switching.

Advanced-T margin to switch to R (°C) — Hysteresis to return from T correction to R correction. Improves switching stability between channels.

Advanced-|Tact–Ttgt| max °C — Maximum allowed difference between actual and target temperature to enable predictive correction. If exceeded, the predictor is inhibited to avoid correcting in unstable conditions.

Advanced-T settle N— Predictor window (number of samples per cycle). Higher N filters noise and stabilizes; lower N speeds up corrections but may overfit. Suggested starting point: N=3.

Advanced-CYCLE — CYCLE — Telemetry and predictor cycle period, in seconds. Allowed values: 5 or 10 (default 10).

Defines how often the MK3 closes a sampling window and performs a diameter predictor evaluation.

With 10 s, the system filters noise better and corrections are more stable.

With 5 s, it reacts faster to diameter changes, at the cost of greater sensitivity to sudden variations.

Advanced-Med jump mm (DIAMJDB) — How much the ~10 s rolling median diameter must change versus the last published median to treat it as a “large jump” that is NOT applied immediately; the MK3 keeps the previous published median while awaiting confirmation in thefollowing window before updating predictor logic and CSV. Higher values tolerate transients better (e.g. inserting/cutting filament); lower values react faster but trigger debounce more often. It does not fix systematic sensor calibration — use DOFF/Hall for bias.

Advanced-Med confirm mm (DIAMJPM) — After a pending large jump crosses "Med jump", firmware requires that the raw median of the next window matches the pending candidate within ± this tolerance (absolute) before publishing; otherwise the hold can continue/reject. It is the confirmation slack between successive windows: higher accepts more easily; lower demands a cleaner match.

Advanced-DIAMDEBUG — MK3 CSV format only: appends lightweight diagnostic columns to see FIFO floor rejects (corr_drop_n), median debounce holding (corr_jump_hold, jump_pending_mm), and raw vs accepted FIFO fill (fifo_n_raw / fifo_n_used). Handy during tuning; commonly OFF during normal runs for shorter rows. Does not alter predictor correction logic. The CSV is send to Python control to process.

Advanced-dE min — Minimum RPM correction step per cycle when acting on R. Prevents corrections that are too small to have practical effect.

Advanced-dE max — Maximum RPM correction step per cycle. Limits abrupt changes to avoid mechanical/flow instability.

Advanced-kSpan E — E gain associated with distance to the diameter band (span component). Higher values respond more strongly when outside band.

Advanced-kErr E — E gain associated with absolute diameter error (error component). Increasing it speeds R correction, but can induce overshoot.

Advanced-dT min C — Desired/base thermal change per cycle when channel T decides to act. Represents the minimum thermal correction intent.

Advanced-dT max C — Maximum allowed temperature change per correction/cycle. In practice: applied dT = min(desired dT, max dT). Keep low (e.g., 1 °C) due to thermal inertia.

Advanced-kSpan T — Thermal gain by distance to band (span component of T channel).

Advanced-kErr T — Thermal gain by diameter error (error component of T channel). High values correct faster but may cause thermal oscillation.

4.2.1 Recommended Settings

Diamdebug: ONModo Manual/autiomatic

Traget Diam: 1.75

Deadband/2: 0.03

Med Salto m: 0.15

Med conf mm:0.06

Tem match: 2

E min:9

E max: 32

T min :174

Tmax:189

CYCLE: 10

dE min: 0.05

dE max:0.6

kSpan E: 10

kErr E:15

dT min:0.2

dT max:1

kSpan T:15

kErr T:55

E margin: 2

T margin: 2

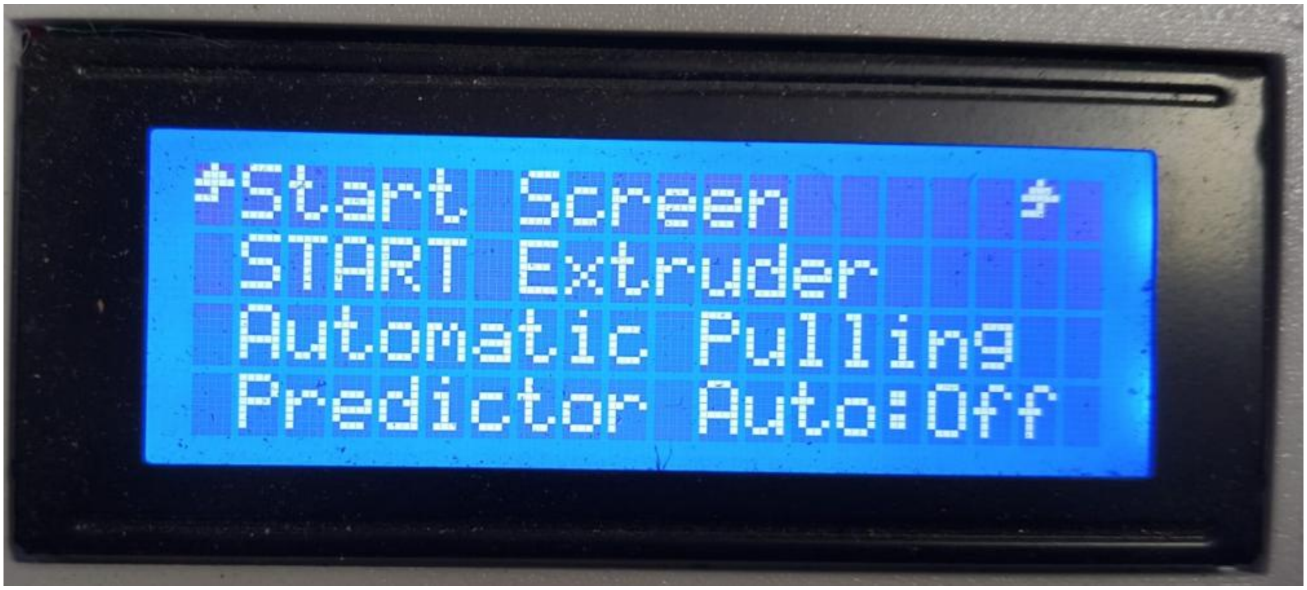
T settle N: 3

Window N: 3

5. Operation recommendations

Objectives

Steps and recommendations for extrusion with automatic predictive diameter control (MK3S+ FR3D Addon, standalone LCD).



Step 1 — Prepare (Predictor Auto OFF)

• Set Predictor Auto: OFF.

• Review diameter-sensor calibration (Hall A3 or ARTME 3D optical).

• Place shredded material in the hopper.

• Make sure the material is dry.

Step 2 — Base setpoints

Go to Settings and assign:

• Temperature

• Extruder RPM

• Fan

Step 3 — Start extrusion

Once the target temperature is reached, start extrusion.

Step 4 — Thread and enable pulling

Thread the extruded filament and set Pulling (spooler) to Automatic.

Step 5 — Enable Predictor Auto

When automatic pulling is stable and the diameter on the LCD is close to the target (shown as Dxxxx / average), set Predictor Auto: ON.

From this point, the prediction algorithm applies automatic diameter corrections each operating cycle.

Step 6 — Predictor reaches temperature / RPM limits

If the predictor saturates against its configured temperature and RPM limits (Tmin/Tmax, Emin/Emax), stop relying on further predictive corrections and apply the operator tips below (from in-app Help: Tips & tricks).

Filament too thin

When average diameter (Dprom) is below the acceptable window, work through these steps in order:

• Nozzle FAN — Manually increase nozzle fan speed (F) so the filament stretches less under tension before solidifying.

• Mechanical distance adjustment — Manually decrease the physical sensor distance (S), moving it closer to the nozzle to give the molten plastic less stretch time.

• Physical check and calibration — Inspect internal wear of the nozzle orifice. Verify the diameter sensor is calibrated and not reporting false readings.

• Batch recipe adjustment — Decrease recycled material percentage (-T%) and increase virgin material percentage (+V%) to reduce melt fluidity.

Filament too thick

When average diameter (Dprom) is above the acceptable window, work through these steps in order:

• Nozzle FAN — Manually reduce nozzle fan speed (F) so the filament can stretch under tension before solidifying.

• Mechanical distance adjustment — Manually increase the physical sensor distance (S), moving it away from the nozzle to give the molten plastic more stretch time.

• Physical check and calibration — Inspect internal wear of the nozzle orifice. Verify the diameter sensor is calibrated and not reporting false readings.

• Batch recipe adjustment — Increase recycled material percentage (+T%) and decrease virgin material percentage (-V%) to increase melt fluidity.